



SNAP RESEARCH

AI RESEARCH ASSISTANT

The Problem

Information Overload: Standard search engines return massive, unorganized lists of links, many times the actual content can be little fade away

Conversational Fatigue: Standard AI tools give conversational, unstructured prose instead of structured data., if it has given structured data also it can be not in a chronological order

Depth Control: No clean way to toggle between basic summaries and expert-level code/mechanisms. Many reports get mixed within then you don't know which is the actual information necessary for you

One Search For Everything : Designed different searches as per what you require for Comparison or Deep dive in any topic

Market Need

Target Users: Students, academic researchers, and developers requiring fast, structured knowledge, St

The Gap: A tool that sits between generic search engines (Google) and deep paper databases (arXiv).and can give a structured answer

The Solution: A lightweight desktop companion providing structured summaries, bookmarks, and fast exports having search option to opt for the accurate one you need to find your solution effectively and in a illustrative way

Foundational Research

There is not a species Foundation Research that has given the exact information that could be use as direct here but there is a bunch of information that to be viewed and reviewed to make make anything regarding the topic

Core Paradigms: Built on Chain-of-Thought (CoT) and Zero-Shot JSON schemas.

UI Guidelines: Designed around Microsoft's Guidelines for Human-AI Interaction (visibility of state, context-aware chips).

Reliability: Prompts researched to force 100% JSON-compliant formatting.

System Architecture

Frontend: HTML5, CSS3, and Vanilla JavaScript (neural canvas animation, session management).

Backend: Flask proxy server (hides API keys, prevents client-side leaks).

AI Engine: Google Gemini API integration via Google AI Studio.

Model Selection & Configuration

Model: gemini-3.5-flash (selected for speed, cost, and high JSON-instruction compliance).

Parameters: Temperature: 0.7 (balanced creativity & factual density), Max Output Tokens: 2048.

Core Feature Demonstration

Explain Mode: Simple educational breakdowns.

Deep Dive Mode: Complex technical mechanisms and definitions.

Compare Mode: Side-by-side matrices outlining use cases and trade-offs.

Synthesize Mode: Linking cross-domain concepts.

Interactivity: Clickable concept tags for auto-search and follow-up chips.

Technical Results

Latency: Low average query response time of 2.8 seconds.

Reliability: 97.3% JSON structural parsing success rate.

Footprint: Very lightweight frontend consuming under 12MB of RAM.

Exporting: Outputs fully clean, parse-ready Markdown files.

Technical Challenges & Solves

Challenge: JSON parsing failures from markdown formatting block fences (``json).

- *Solve:* Built a regex sanitization layer in Python to strip fences before parsing.

Challenge: Unicode character crashes in Windows consoles.

- *Solve:* Replaced unicode status checkmarks with robust ASCII text blocks.

Challenge: Port conflicts (conflict with Other project which was port to 5001)

- *Solve:* Assigned SNAP Research permanently to port 5001 with a port-clearing launcher script (run.bat).

Future Roadmap

As this is simple and basic version of AI research Assistant but to make it more effective and better then the first version it requires , its still in development and betterment phase

Phase 1: Real-source integration

Phase 2: PDF report generator with automated academic citations.

Phase 3: Visual concept mapping (dynamic mind map graphics of research paths).